Implementation of half-subtractor and full subtractor

Objective:

The function of half-subtractor and full subtractor

Theory

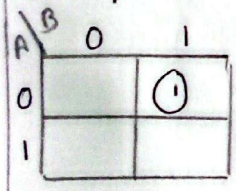
1) Half-subtractor

A combinational circuit that performs subtraction of two bits is called half subtractor. This circuit need two binary inputs and two outputs. The output variable produce difference and borrow.

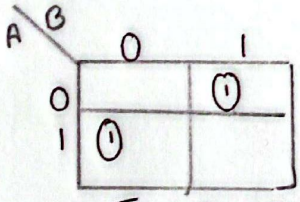
Truth-table

A	B	Borrow	Diff
0	0	0	0
0	1	1	1
1	0	0	1
1	1	0	0

K-map



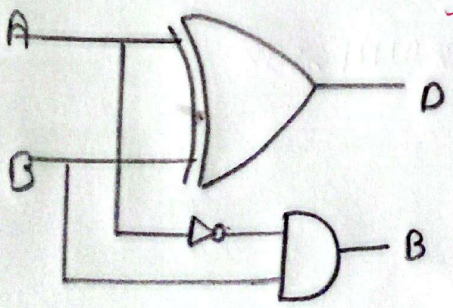
$B = A'B$



$F = A'B + AB'$

$= A \oplus B$

Implementation using X-OR and AND Gate



2) Full-subtractor

A combinational circuit that performs the subtraction of three bits is call full-subtractor.

Truth table

A	B	C	Borrow	difference
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	1	0
1	0	0	0	1
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

K-map

A \ BC	00	01	11	10
0		1	1	1
1			1	

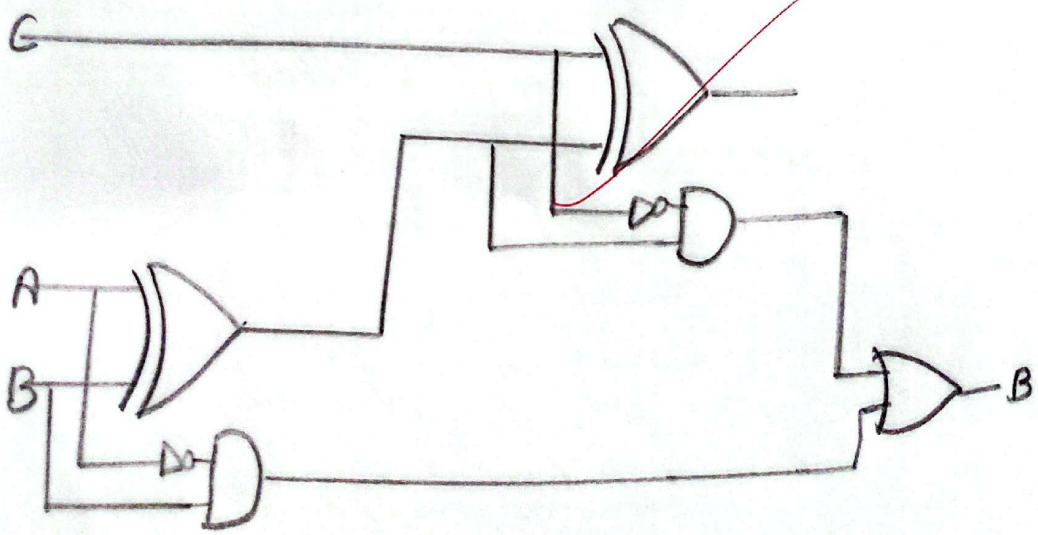
$F = \bar{A}B + \bar{A}C + BC$

A \ BC	00	01	11	10
0		1		1
1	1		1	

$F = \bar{A}B'C + \bar{A}B\bar{C} + ABC$

$= \bar{A}C \oplus (A \oplus C)$

Implementation of full subtractor



Observation

1) Implementation and verification

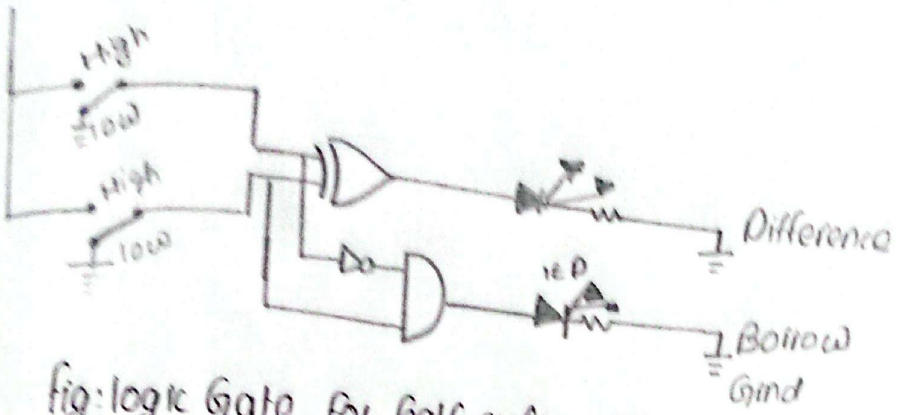


fig: logic Gate for half-subtractor

Inputs		Expected output		Observed output	
A	B	Borrow	diff	Borrow	diff
0	0	0	0	0	0
0	1	1	1	1	1
1	0	0	1	0	1
1	1	0	0	0	0

Observed output and expected output are same

2) Implementation & verification of full subtractor

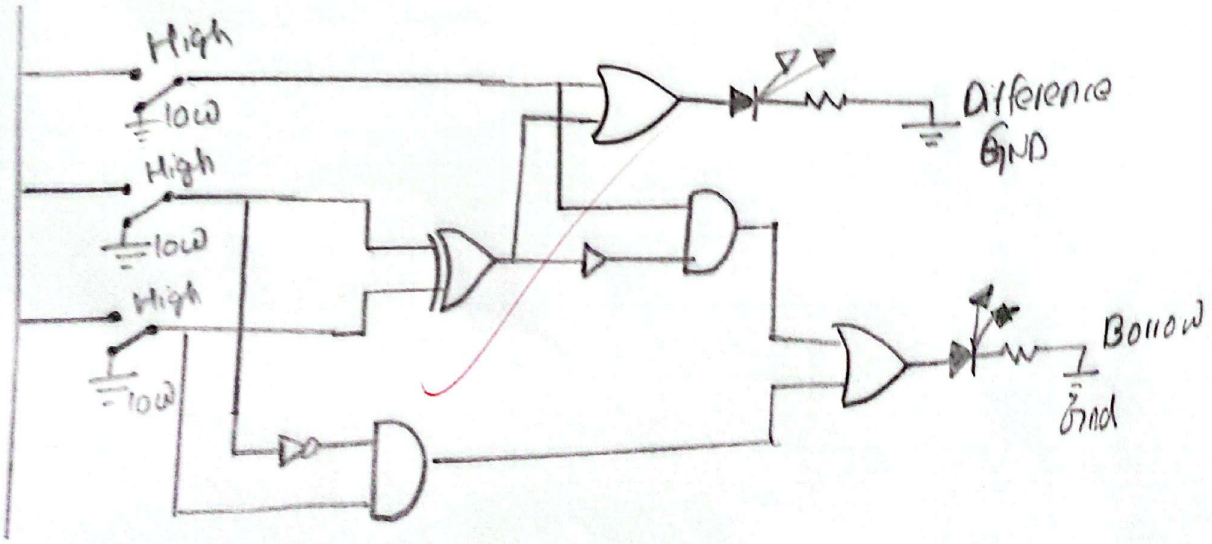


fig: logic Gate for full subtractor.

Inputs			Expected Out		observed O	
A	B	C	B	O	B	D
0	0	0	0	0	0	0
0	0	1	1	1	1	1
0	1	0	1	1	1	1
0	1	1	1	0	1	0
1	0	0	0	1	0	1
1	0	1	0	0	0	0
1	1	0	0	0	0	0
1	1	1	1	1	1	1

Conclusion.

In this way, we learnt about the implementation of half-subtractor and full-subtractor. we learnt that we get two output one of difference and one for borrow.

CamScanner